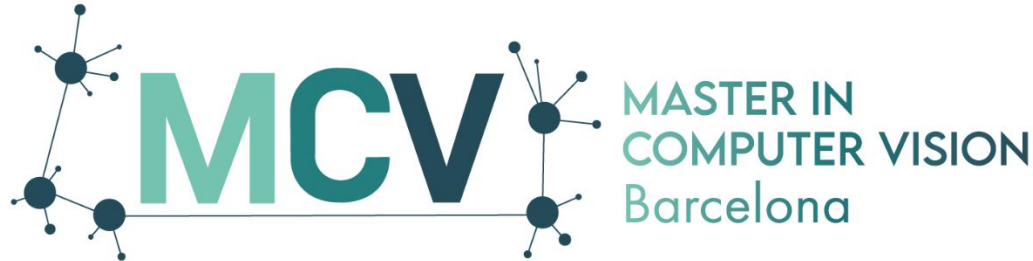


C6: Multi-camera tracking

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Team 4 members

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Pau Monserrat



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Problem Statement

Problem statement

Multi-Target Multi-Camera (MTMC) tracking aims to detect, track, and re-identify vehicles across video feeds from multiple cameras, assigning a consistent **global ID** to each vehicle throughout the camera network.

Our goal: evaluate a MTMC tracking pipeline on the **CVPR 2022 AI City Challenge** [1], using **IDF1** as the main metric.

Key challenges:

Appearance variation

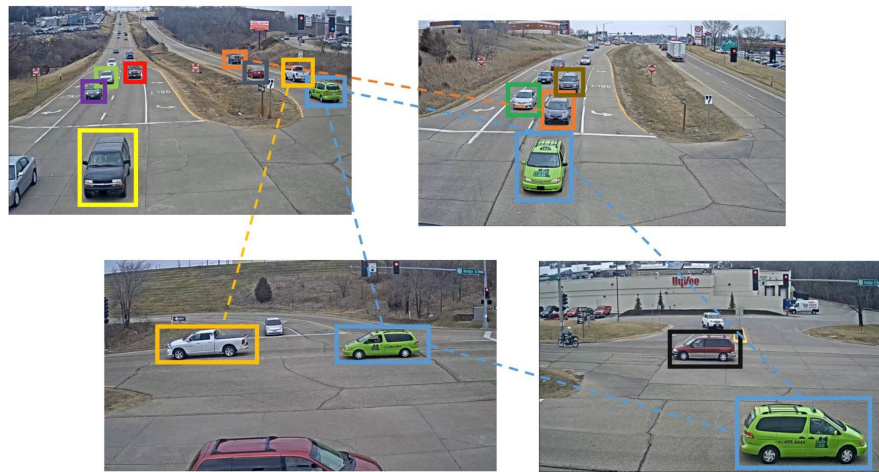
Same vehicle looks different across cameras due to viewpoint, lighting, and occlusion.

Spatial-temporal reasoning

Vehicles must be matched considering plausible travel times and camera topology.

Scale

CityFlow spans up to 40 cameras and 666 unique IDs across 5 scenarios.



[1] Tang, Z., Naphade, M., Liu, M.-Y., Yang, X., Birchfield, S., Wang, S., Kumar, R., Anastasiu, D., & Hwang, J.-N. (2019). CityFlow: A City-Scale Benchmark for Multi-Target Multi-Camera Vehicle Tracking and Re-Identification. CVPR 2019.

Pipeline

Pipeline

Our implementation follows an **offline** approach: all cameras are processed sequentially, and cross-camera matching is performed **after** all single-camera tracks have been extracted.



1. Car Detection

We use a detector to predict the bounding boxes of the cars in each frame. **YOLOv10s**.

2. Single-Camera Tracking

With the extracted bounding boxes, we perform a tracking-by-detection algorithm for every single camera. **SORT**.

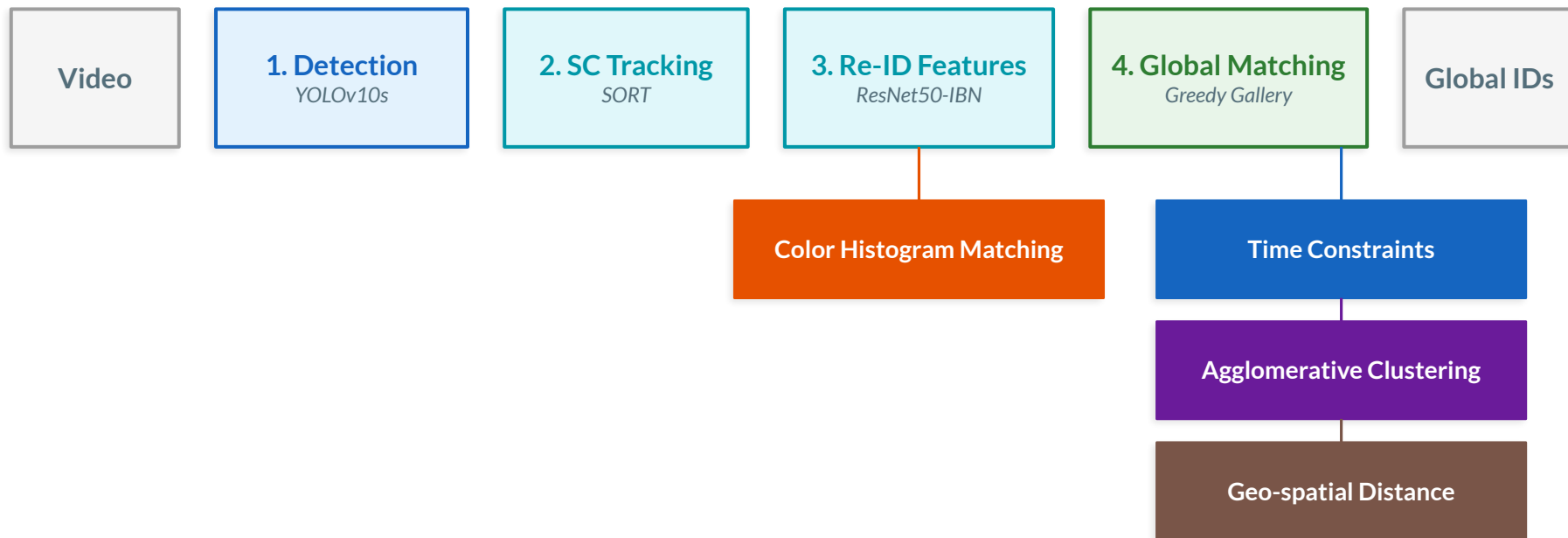
3. Re-ID Features

We extract a signature for each track using a Feature Extraction Backbone. **ResNet50-IBN**.

4. Global Matching

Similar signatures are associated with the same global ID. **Greedy matching**.

Pipeline - Additional Techniques



Basic Techniques

Detection

Last week we saw that our detector was **not robust** enough → Overfitted to c010 sequence.

We have trained a **YOLOv10s** on COCO (keeping only images where *car* or *truck* appear) and first 25% of c010 camera.

Previous

Erratic bounding boxes. Detections lost in a lot of frames.



New

Tight bounding boxes. Detections lost just in few frames.



We filter the detections using the ROI. Those falling partially or totally inside the red area are filtered.

Single-Camera Tracking

For each camera of the sequence, we use **SORT** to obtain a history of the local tracks.

From past weeks, we have seen that SORT is the best in **keeping identities** (the one with less ID switches).



Hyperparameter	Value
IoU threshold	0.2
Max age	12 frames
Confidence threshold	0.65

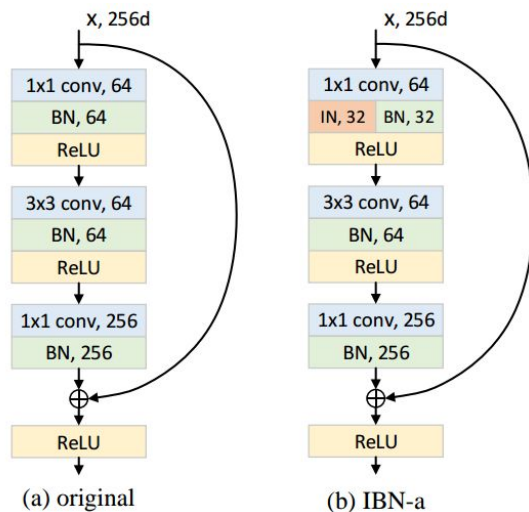
Tracks are finalized once they leave the ROI or exceed the maximum age.

Feature Embedding

After obtaining the local tracks of each camera, we compute the **signature** for each of them.

We use a **ResNet50-IBN** [2] trained for vehicle re-identification [3].

This architecture integrates **Instance Normalization (IN)** with **Batch Normalization (BN)**.



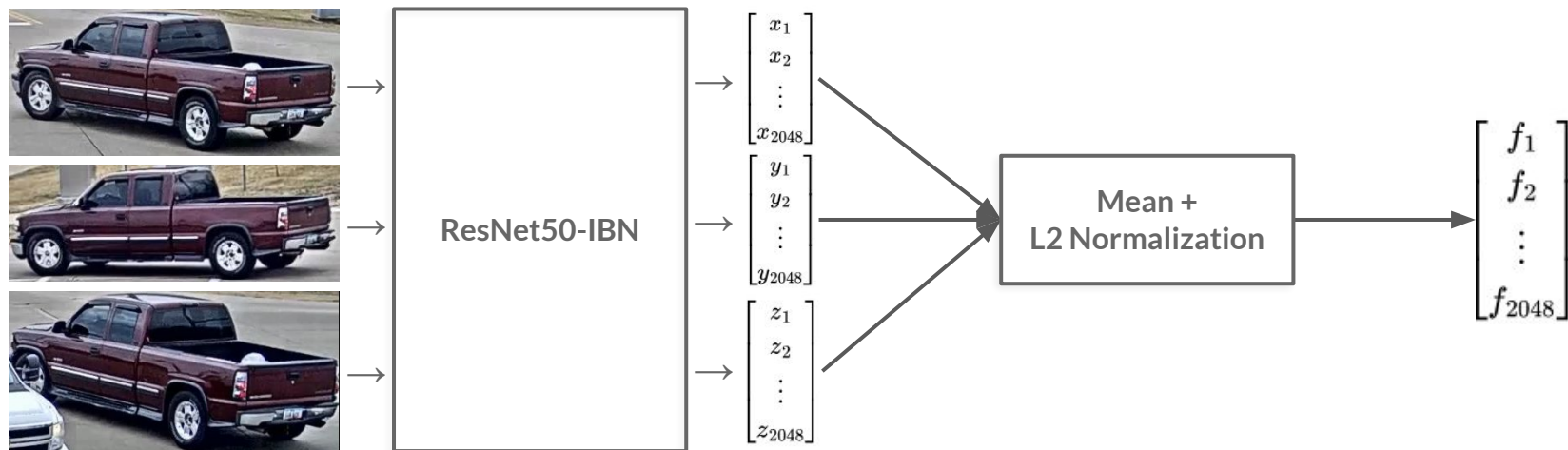
Key insights:

- IN normalizes each image independently, erasing lighting, contrast and color information. **Content focus.**
- BN normalizes across the whole batch, preserving the discriminative features. **Style focus.**
- 50/50 channel split allows the network to learn a **color-invariant** but **identity-aware** representation.

[2] Xingang Pan, & Xiaoou Tang (2018). Two at Once: Enhancing Learning and Generalization Capacities via IBN-Net. ECCV. ([Github](#))

[3] Szűcs, G., Borsodi, R., Papp, D. (2023). Multi-Camera Trajectory Matching based on Hierarchical Clustering and Constraints. Multimedia Tools and Applications. ([Github](#))

Feature Embedding

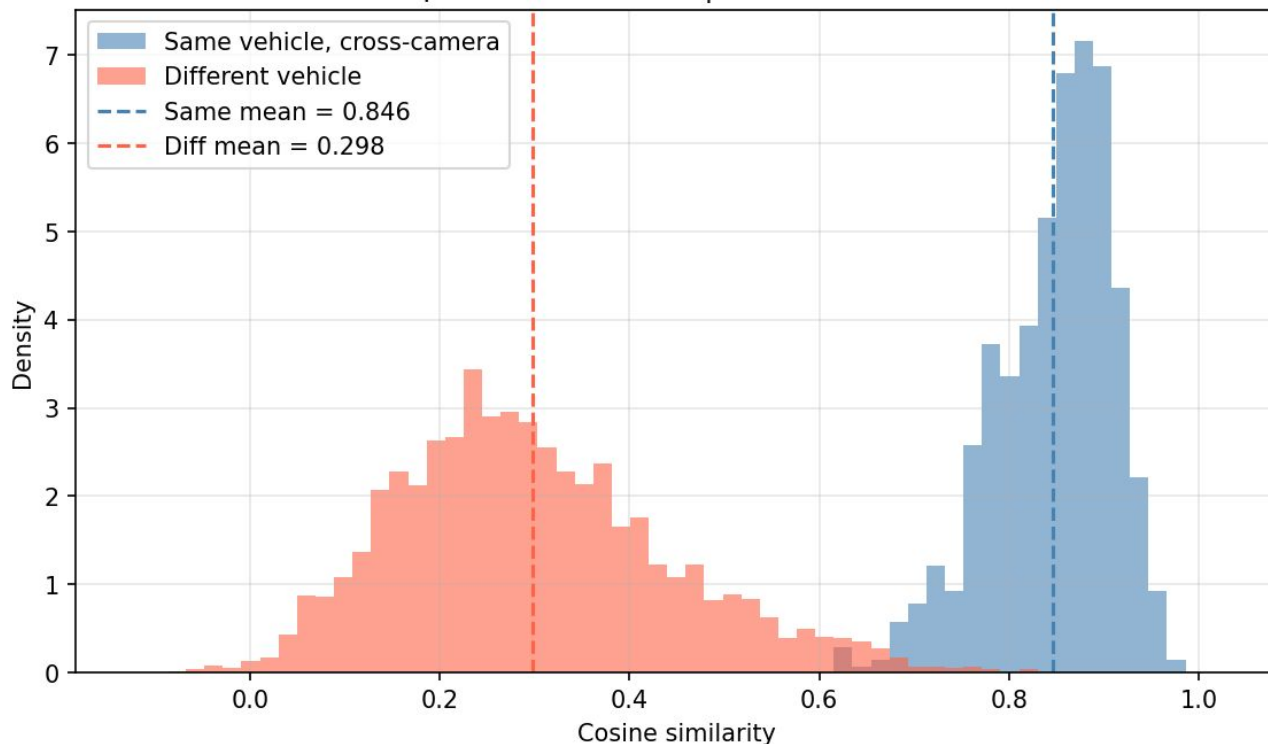


N crops are picked from the entire track (hyperparameter).

The final signature vector has 2048 dimensions. They are compared using **cosine similarity**.

Feature Embedding

ReID feature similarity distributions — S01
Separation = 0.5489 | Rank-1 = 89.4%



Experiment done using GT tracks:

1. Extract signature of each track.
2. **Pairwise** CS calculation.

If the tracks are perfect, Re-ID works incredibly well.

ID switches corrupt signatures.

Base Results

With all this set, we perform a **greedy matching**:

1. **Global gallery** is created (each global ID has a signature).
2. For each track:
 - a. Signature **compared** against all entries.
 - b. Best match higher than the threshold → the track is **associated** with that global ID.
 - c. Score below the threshold → new global ID **created**.

Signature update: for a found match, the global gallery's signature is updated using a **momentum average** (80% old signature, 20% new).

Hyperparameter	Value
IoU threshold	0.5
Match threshold	0.68
Max age	12 frames
Confidence threshold	0.65
Num crops per track **	5

*Hyperparameters found doing a Bayes search on SEQ 1

** Set to 5 for all the experiments

	IDF1	HOTA
SEQ 1	37.26	27.14
SEQ 3	14.37	15.62
SEQ 4	31.04	24.90
Average	27.56	22.55

We try to enhance these metrics with the addition of new techniques.

Additional Techniques

Time Constraints

We decided to apply two main **time-related constraints**:

A car cannot appear in the same camera twice

If two cameras point at the same area, a vehicle can't reappear after X^* seconds

*Given hyperparameter search X was set to 6

S01

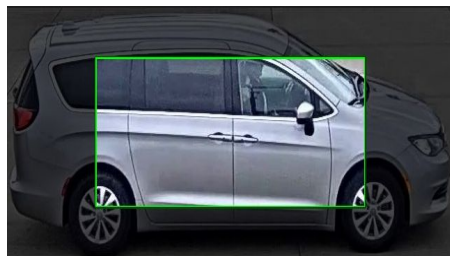
Method	IDF1	HOTA
Baseline ReID	37.26	27.14
No Repetition + ReID	43.18	30.53
Reappearance Limit + ReID	44.48	31.69
Both Constraints + ReID	43.24	30.61

Average

Method	IDF1	HOTA
Baseline ReID	27.56	22.55
No Repetition + ReID	29.53	23.68
Reappearance Limit + ReID	29.96	24.07
Both Constraints + ReID	29.55	23.71

Color Histogram Matching

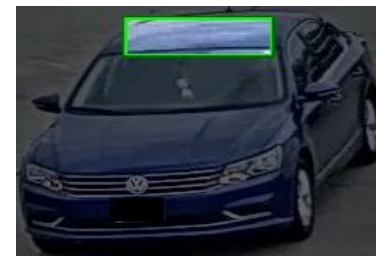
Central Region



Hypothesis

- Remove **non-car noise**
- Focus on the car's **paint colour**.
- **Problem:** Windows and car lights not heavily identifiable

Roof Region



Hypothesis

- Avoids some **not heavily identifiable objects**.
- **Problem:** Roof has more reflections
- **Problem:** Heavily dependant on good camera position.

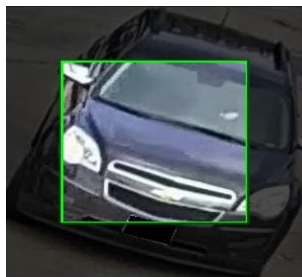
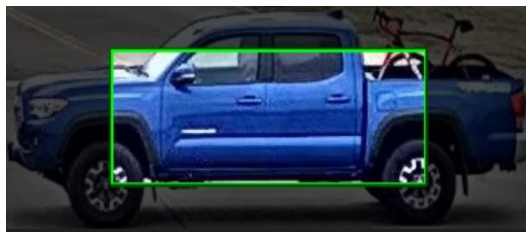
We then converted these areas to:

- **Full HSV histograms**
- **Hue-Only HSV histograms.**

Color Histogram Matching

Reality

Central Region



Roof Region



Average

Method	IDF1	HOTA
Time-Constrained ReID	29.96	24.07
Central Region - HSV	19.68	17.30
Central Region - Hue	24.63	19.54
Roof Region - HSV	21.57	18.28
Roof Region - Hue	23.90	19.65

BBox Shape Information

To add BBox shape information into the equation, we used the following values:

Aspect Ratio

Width / Height

Encodes shape of the vehicle

More homogenously shaped vehicles
get a value closer to 1

Size Change Rate

Area of last frame - Area of current
frame

Encodes Speed

Faster vehicles have a higher rate.

Average

Method	IDF1	HOTA
Time-Constrained ReID	29.96	24.07
BBox Shape + ReID	28.54	22.76
Color + BBox Shape + ReID	19.28	16.41

Agglomerative Clustering

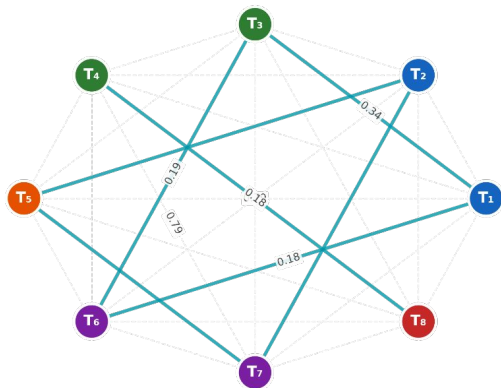
We build a **fully-connected graph** where each node is a single-camera tracklet and each edge carries an **appearance cost** (cosine distance between Re-ID embeddings).

Agglomerative clustering groups low-cost nodes into clusters → each cluster = one global vehicle ID.

1. Build the cost matrix

Cosine distance between every pair of tracklets (across cameras) → symmetric $N \times N$ distance matrix.

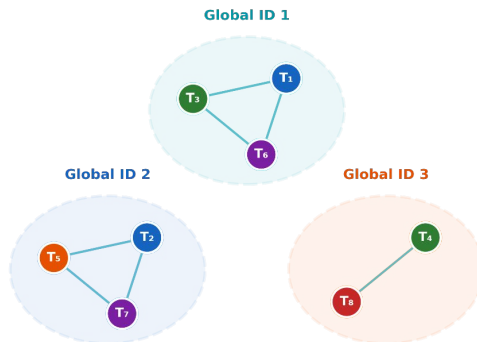
Fully-connected appearance graph



2. Agglomerative clustering

Each tracklet starts as its own cluster. Iteratively merge closest two until distances exceed threshold.

After agglomerative clustering



3. Assign global IDs

Each resulting cluster groups tracklets from multiple cameras = same vehicle → unique global ID.

Edge weight

= cosine distance between Re-ID embeddings

— Solid teal

Low cost (similar appearance)

--- Dashed gray

High cost (different vehicles)

Dashed region

= one global vehicle identity

(tracklets from different cameras grouped together)

Agglomerative Clustering: Hyperparameter tuning

We sweep three hyperparameters of the agglomerative clustering to maximise **IDF1** across all sequences.

Hyperparameters to tune:

- Distance threshold**

Max cosine distance to merge clusters

- Linkage**

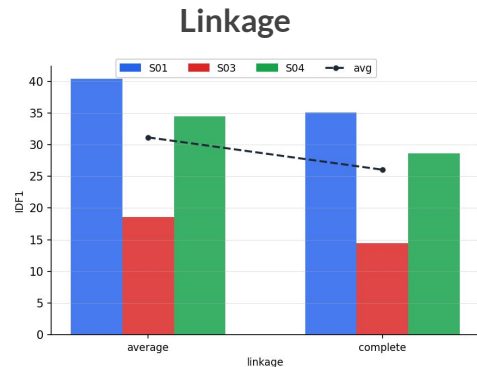
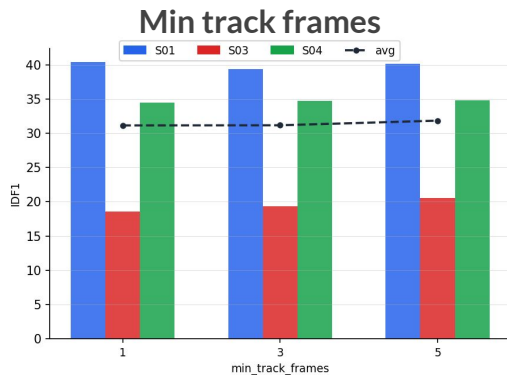
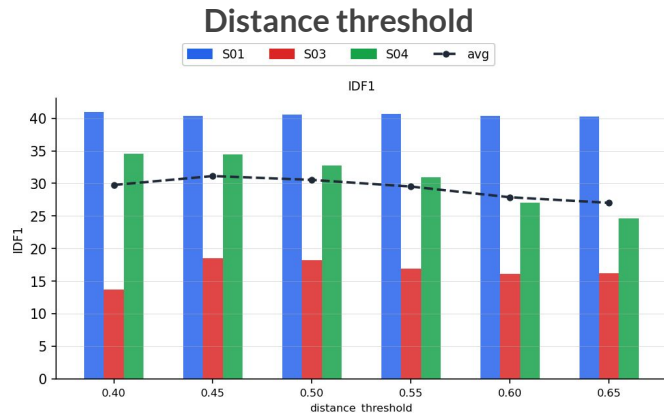
Average vs complete linkage

- Min track frames**

Minimum tracklet length to consider

Best config

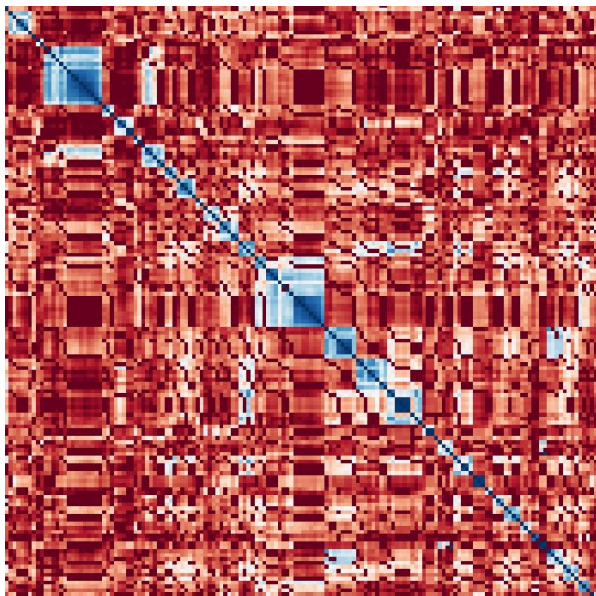
- Distance threshold → 0.45
- Linkage → average
- Min track frames → 5



Agglomerative Clustering: Results

Best config: distance_threshold = 0.45 · linkage = average · min_track_frames = 5

Visible clusters on the cost matrix



	IDF1	HOTA
SEQ 1	40.15	29.43
SEQ 3	20.58	20.71
SEQ 4	34.77	26.26
Average	31.83	25.47

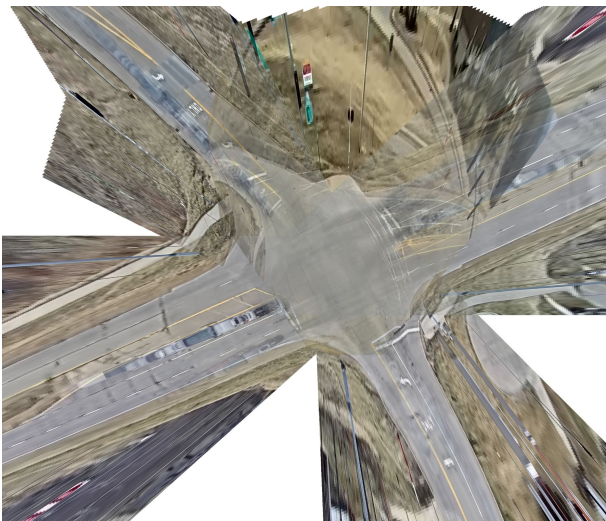
Best average performance so far.

Key takeaways

- Sequence 1 is **very stable** when changing the threshold
- Average is better
- Discarding short tracklets **barely affects performance**
- Does not improve each sequence, but **better in average**

World-Camera Position Information

The ground truth provides homographies that map GPS coordinates to camera pixels → we can exploit world space data



When projecting the cameras to world space, we can see two issues:

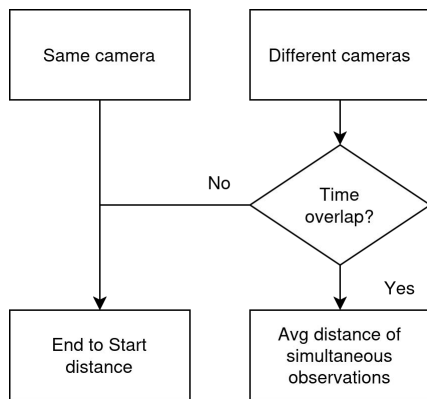
- Points **do not overlap** correctly.
- The further away, **the less certainty**.

World-Camera Position Information

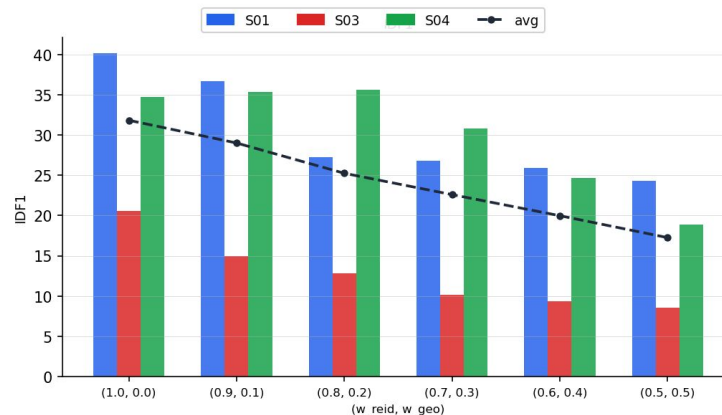
Idea: use a distance cost when matching tracklets

How can we measure distance between tracks?

Distance computation



$$C = w_a \cdot \text{appearance_distance} + w_g \cdot \text{geometric_distance}$$



Key takeaways

- Adding a distance cost decreases performance.

Reason: reprojection errors (calibration and noisy bboxes).

- Geo-spatial information does not aid clustering



Conclusions

Conclusions

- **YOLOv10s** as car detector given last weeks' performance.
- Single Camera Tracking with **SORT** using Hungarian IoU assignment
- **Pretrained ResNet** as feature extractor.
- **Greedy Sequential Gallery** for global matching between cameras.

Key Takeaways

- Time restrictions work well, but depend on **prior knowledge**. With more time it could be automated.
- Color histograms are heavily sensible to different camera positions.
- Color information is **already encoded** by the ResNet, thus the diminishing results.
- Reality is that Single-Camera Tracking **depends heavily** in Re-ID matching.
- Using geospatial information with the chosen clustering algorithm **does not benefit**.